

Fourier Decay of Rvachev's Up-Function

Final synthesis of pointwise, metric, mean, extremal, and annular asymptotics

Research article for Vladimir Reshetnikov's ProveIt project

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Abstract

Let

$$\Phi(x) = \widehat{\text{up}}(x) = \prod_{h=0}^{\infty} \text{sinc}\left(\frac{\pi x}{2^h}\right), \quad \text{sinc } z = \frac{\sin z}{z},$$

be the Fourier transform of Rvachev's compactly supported up-function. This article audits four independent first-wave analyses, the relevant lacunary-products paper of Aistleitner–Hofer–Larcher, and the subsequent comparative audit, and then gives a single self-contained synthesis. The main conclusion is that the phrase “the decay rate” has no unqualified meaning. There is a universal deterministic factor

$$\exp\left(-\frac{(\log x)^2}{2 \log 2}\right),$$

but the next power of x depends on whether amplitude means a typical value, an arithmetic mean, an RMS value, a pointwise upper envelope, or the largest value in a dyadic annulus.

An exact dyadic factorization reduces the remaining numerator to a lacunary sine cocycle for the doubling map. Its L^q pressure yields a complete shell spectrum. In particular, with $E_{\kappa}(x) = x^{-\kappa} \exp(-(\log x)^2/(2 \log 2))$, the four principal powers are

$$\begin{aligned} \kappa_0 &= 3.151496129472319\dots, & \kappa_1 &= 2.748070014871334\dots, \\ \kappa_2 &= 2.651496129472319\dots, & \kappa_{\infty} &= 2.359014879111741\dots \end{aligned}$$

They correspond respectively to geometric mean / almost-everywhere dyadic rays, arithmetic mean, root mean square, and the sharp pointwise upper envelope. The old conjectural power $\log_2(2\pi)$ is therefore exactly right for RMS decay and wrong for the other readings. No universal power of $\log x$ repairs the mismatch.

The metric fluctuation is also settled. For $S_n(t) = \sum_{j < n} \log |2 \sin(\pi 2^j t)|$, an exact covariance computation gives

$$\int_0^1 S_n(t)^2 dt = \frac{\pi^2}{4}n - \frac{\pi^2}{3}(1 - 2^{-n}).$$

Equivalently, S_n is a coboundary perturbation of the reverse martingale-difference sequence generated by $\log |\tan \pi t|$. Hence the correct law-of-the-iterated-logarithm constant is $\pi/2$ with denominator $\sqrt{2n \log \log n}$, or $\pi/\sqrt{2}$ with denominator $\sqrt{n \log \log n}$. The factor π printed in the 2017 lacunary-products theorem is too large by $\sqrt{2}$; the source of the discrepancy is the missing $1/2$ in the L^2 norm of the cosine basis.

For the literal Cesaro question, the arithmetic-mean gauge $g(x) = A_1 E_{\kappa_1}(x)$, where $A_1 = 0.091266124131588\dots$, gives shell means tending to one and solves the requested two-sided boundedness condition. It also gives convergence at dyadic endpoints. At unrestricted endpoints a nonconstant dyadic-mantissa profile remains, governed by a continuous singular eigenmeasure of an explicit transfer operator. We prove that no gauge with a stabilizing within-shell shape (including powers and iterated logarithms, regularly varying factors, and fixed log-periodic corrections) can remove this profile. A completely unrestricted exotic scale-dependent analytic gauge is not ruled out by that theorem. The multiplicatively natural logarithmic Cesaro mean does converge for all endpoints, with normalizing constant $A_1^{\log} = 0.09386063575678\dots$

Finally, the raw annular maximum $\mathfrak{M}_k = \max_{2^k \leq x \leq 2^{k+1}} |\Phi(x)|$ has a sharper boundary-layer law:

$$\log \mathfrak{M}_k = -\frac{\log 2}{2} k^2 - (\log 2) \kappa_\infty k - \frac{W(c(k+1))^2 + 2W(c(k+1))}{2 \log 2} + O((\log \log k)^2),$$

where $c = (\log 2) \pi^{-1} \sqrt{3/2}$ and W is the principal Lambert function. Thus the largest point in a shell approaches its left boundary and carries a second log-normal factor on the $\log \log x$ scale. In contrast, the unique lobe maxima immediately after powers of two form explicit deep valleys with leading coefficient $-1/\log 2$, twice the generic log-square coefficient. Together these results reconcile every substantive disagreement among the source documents and give the final rigorous decay dictionary.

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1 The question and the shape of the answer

Rvachev's up-function is the even C^∞ probability density supported on $[-1, 1]$ that is closely related to the Fabius distribution function. Under the Fourier convention

$$\widehat{f}(\xi) = \int_{\mathbb{R}} f(t) e^{-2\pi i t \xi} dt,$$

its transform is the entire function

$$\Phi(z) = \widehat{\text{up}}(z) = \prod_{h=0}^{\infty} \text{sinc}\left(\frac{\pi z}{2^h}\right). \quad (1)$$

This formula is classical; see, for example, Arias de Reyna's systematic treatment [1] and the Fabius/Rvachev development in [11].

The motivating questions in [9, 10] ask for the amplitude of the oscillations and, more specifically, for a positive elementary function g such that an average of $|\Phi|/g$ tends to one or at least stays between two positive constants. A working draft in the `ProveIt` repository [12] proposes a log-normal factor together with the power $x^{-\log(2\pi)/\log 2}$ and a possible power of $\log x$. The log-normal factor is indeed fundamental, but the rest of the conclusion depends on which norm or averaging operation is intended.

There are three separate obstructions to a single pointwise answer.

1. The transform has a zero at every nonzero integer, so no asymptotic $|\Phi(x)| \sim g(x) > 0$ can hold on the full real ray.
2. After the deterministic denominator is removed, what remains is a Birkhoff product for the doubling map. Its geometric mean, arithmetic mean, quadratic mean, and supremum have different exponential growth rates.
3. Even the unique maxima between consecutive zeros possess exceptional subsequences. The peaks associated with the binary period-two orbit attain the sharp global upper envelope, whereas the extrema in the first few intervals after a power of two are exponentially smaller on the n^2 scale.

Thus the correct answer is a *spectrum of decay rates*, not one formula. A convenient common scale is

$$E_\kappa(x) := \exp\left(-\frac{(\log x)^2}{2a} - \kappa \log x\right) = x^{-\kappa} \exp\left(-\frac{(\log x)^2}{2 \log 2}\right), \quad a := \log 2. \quad (2)$$

For fixed dyadic phase, the quadratic term in (2) comes from the product of the n denominators $2^1, \dots, 2^n$. The linear term is where the lacunary numerator enters.

1.1 Main decay table

The following table summarizes the results proved below. In the finite- q rows, “normalized” means the normalized L^q norm on the dyadic shell $[2^{n-1}, 2^n]$.

Amplitude notion	numerator scale Λ	power κ in E_κ	conclusion
Geometric mean ($q = 0$) and typical dyadic ray	$1/2$	$\frac{3}{2} + \frac{\log \pi}{a}$	shell geometric mean is exactly constant after normalization; the same exponent holds for Lebesgue-a.e. fixed phase
Arithmetic mean ($q = 1$)	$\rho_1 = 0.6613226021 \dots$	$\frac{\log \pi + a/2 - \log \rho_1}{a}$	shell means converge to a positive constant; ordinary Cesaro means have a nonconstant dyadic phase profile
Root mean square ($q = 2$)	$2^{-1/2}$	$\frac{\log(2\pi)}{a}$	exact numerator identity $\int B_n^2 = 2^{-n}$; normalized shell RMS converges
Uniform envelope ($q = \infty$)	$\sqrt{3}/2$	$\frac{3}{2} + \frac{\log(\pi/\sqrt{3})}{a}$	global upper bound, sharp on $x = 2^n(2/3)$; not a two-sided envelope for every extremum

The ordering is

$$\begin{aligned} \kappa_\infty = 2.359014879 \dots < \kappa_2 = 2.651496129 \dots \\ < \kappa_1 = 2.748070014 \dots < \kappa_0 = 3.151496129 \dots \end{aligned} \quad (3)$$

Larger q emphasizes rarer and larger values, hence gives a slower decay and a smaller power κ_q .

The coefficient $\log(2\pi)/\log 2$ appearing in the working draft is not an accidental near miss. It is *exactly* the root-mean-square power κ_2 . It is not the arithmetic-mean power, the typical power, or the sharp uniform-envelope power. No factor $(\log x)^p$ can convert one of these powers into another, because their discrepancy is already a nonzero multiple of $\log x$ in the logarithm of the amplitude.

2 Audit map and final verdict

The source corpus contains four first-wave documents and one second-wave comparative audit. Their apparent disagreement is mostly a disagreement about which functional of $|\Phi|$ is being estimated. The following table records the final verdict before the detailed proofs.

Source	Main contribution	Final assessment
Wave 1, Document 1 [19]	Exact dyadic reduction, typical law, martingale CLT/LIL, and a sharp pointwise envelope.	Substantively correct on these layers. Its LIL constant $\pi/\sqrt{2}$ is the correct one; the argument is completed in Section 7 .
Wave 1, Document 2 [20]	Sharp raw dyadic-annulus maxima, Lambert- W boundary layer, and explicit dyadic troughs.	Correct and strictly sharper than the other documents on annular maxima. The proof is reconstructed in Section 11 .
Wave 1, Document 3 [21]	Typical, extremal, arithmetic-mean, and L^q interpretations; an analytic Cesaro gauge.	Correct except that it faithfully inherited the published LIL constant π , which is too large by $\sqrt{2}$.
Wave 1, Document 4 [22]	Most systematic transfer-operator treatment, exact RMS identity, sharp envelope, phase profile, and fixed-offset valley theorem.	Correct modulo its explicitly imported Ruelle–Perron–Frobenius block; it omits the fluctuation and Lambert boundary-layer layers rather than contradicting them.
Aistleitner–Hofer–Larcher [15]	Sharp metric study of the same lacunary products and a rigorous five-digit enclosure of the L^1 Perron root.	The order estimates and L^1 input are valid. The printed LIL constant contains a factor- $\sqrt{2}$ normalization error, located precisely in Section 7.3 .
Comparative audit [23]	Reconciles the four readings, locates the LIL error, proves a stabilizing-gauge obstruction, and adds a logarithmic-mean repair.	Essentially correct. Its phrase “every elementary gauge” must be read as the explicitly proved stabilizing/Hardy-field class; no theorem in the corpus excludes every conceivable scale-dependent analytic gauge.

The final answer can therefore be stated compactly.

Theorem 2.1 (Decay dictionary). *Let E_κ be defined by (2).*

1. *For Lebesgue-a.e. dyadic phase, $|\Phi| = E_{\kappa_0} \exp(o(\log x))$, with Gaussian fluctuations of variance $\pi^2/4$ per dyadic level and the corrected LIL of [Theorem 7.2](#).*
2. *The normalized shell L^q means converge to positive constants for every fixed $0 < q < \infty$, with exponent κ_q determined by the Perron root of (21). In particular $q = 1$ answers the literal mean-amplitude question and $q = 2$ gives $\kappa_2 = \log_2(2\pi)$ exactly.*
3. *There is $C < \infty$ such that $|\Phi(x)| \leq CE_{\kappa_\infty}(x)$ for $x \geq 1$, and equality up to one fixed constant holds on the ray $x = 2^n(2/3)$.*
4. *The raw annular maxima have the additional Lambert- W suppression stated in [Theorem 11.3](#); the local lobe maxima have both sharp-envelope subsequences and the fixed-offset dyadic valleys of [Theorem 12.1](#).*
5. *The gauge $A_1 E_{\kappa_1}$ solves the two-sided Cesaro boundedness problem and the dyadic-endpoint limit. The unrestricted ordinary limit fails throughout the stabilizing class of [Theorem 9.7](#); the logarithmic mean of [Theorem 9.8](#) converges without endpoint restrictions.*

3 Exact products, zeros, and scaling laws

3.1 Entire convergence and three product forms

The compact convergence of (1) follows from

$$\operatorname{sinc} w = 1 - \frac{w^2}{6} + \mathcal{O}(w^4), \quad \sum_{h \geq 0} \frac{|z|^2}{4^h} < \infty$$

uniformly for z in a compact set. Hence Φ is even, real on $\mathbb{R}$, entire, and $\Phi(0) = 1$.

Vieta's product $\operatorname{sinc} t = \prod_{j \geq 1} \cos(t/2^j)$ and Euler's sine product give two other useful forms:

$$\Phi(z) = \prod_{j=1}^{\infty} \cos^j\left(\frac{\pi z}{2^j}\right) = \prod_{m=1}^{\infty} \left(1 - \frac{z^2}{m^2}\right)^{1+\nu_2(m)}. \quad (4)$$

Indeed, in the cosine product the factor with denominator 2^j occurs for exactly j pairs of indices. In the Weierstrass product, the integer m is a zero of the sinc factor at level h precisely when $2^h \mid m$, so its total multiplicity is $1 + \nu_2(m)$.

The elementary one-step scaling law is

$$\Phi(2x) = \operatorname{sinc}(2\pi x)\Phi(x), \quad \Phi(x) = \operatorname{sinc}(\pi x)\Phi(x/2). \quad (5)$$

3.2 Zeros and their multiplicities

Proposition 3.1 (Integer zero divisor). *The nonzero zeros of Φ are exactly the nonzero integers. The zero at $m \in \mathbb{Z} \setminus \{0\}$ has multiplicity*

$$d(m) = 1 + \nu_2(|m|).$$

Moreover, if $b \geq 1$ is an integer, $d = 1 + \nu_2(b)$, and $z \uparrow b$, then

$$|\Phi(z)| = b^{-d} \left| \Phi(b/2^d) \right| (b - z)^d (1 + \mathcal{O}(b - z)). \quad (6)$$

Proof. The divisor statement follows immediately from the last product in (4). For the local coefficient, the vanishing factors are those with $0 \leq h \leq \nu_2(b)$. At $z = b$ each has a simple zero and

$$\left| \frac{d}{dz} \operatorname{sinc}\left(\frac{\pi z}{2^h}\right) \right|_{z=b} = \frac{1}{b}.$$

The product of the remaining factors is

$$\prod_{h=d}^{\infty} \left| \operatorname{sinc}\left(\frac{\pi b}{2^h}\right) \right| = \left| \Phi(b/2^d) \right|.$$

Multiplication gives (6). □

3.3 Exact dyadic renormalization

Let

$$T(x) = 2x \pmod{1}, \quad s(x) = |\sin(\pi x)|, \quad x \in [0, 1],$$

and define the lacunary product

$$B_n(y) := \prod_{j=1}^n s(T^j y) = \prod_{j=1}^n |\sin(\pi 2^j y)|. \quad (7)$$

Theorem 3.2 (Exact dyadic factorization). *For every $n \geq 1$ and $y \in [1/2, 1]$,*

$$|\Phi(2^n y)| = |\Phi(y)| \pi^{-n} y^{-n} 2^{-n(n+1)/2} B_n(y). \quad (8)$$

For any real κ , set

$$W_\kappa(y) := |\Phi(y)| y^\kappa \exp\left(\frac{(\log y)^2}{2a}\right). \quad (9)$$

Then the following normalized identity is exact:

$$\frac{|\Phi(2^n y)|}{E_\kappa(2^n y)} = W_\kappa(y) \exp(n(\kappa a - \log \pi - a/2)) B_n(y). \quad (10)$$

Proof. Iterating (5) gives

$$\Phi(2^n y) = \Phi(y) \prod_{j=1}^n \text{sinc}(\pi 2^j y).$$

The product of the denominators is $\pi^n y^n 2^{1+\dots+n} = \pi^n y^n 2^{n(n+1)/2}$, proving (8). Write $L = \log(2^n y) = na + \log y$ and divide (8) by $\exp(-L^2/(2a) - \kappa L)$. The terms $-n \log y$ cancel, leaving (10). $\square$

Equation (10) is the central identity of this article. The entire deterministic decay has been extracted into E_κ . The bounded phase factor W_κ and the dynamical product B_n contain all remaining information.

3.4 The Thue–Morse polynomial

Let $\tau_k = (-1)^{s_2(k)}$, where $s_2(k)$ is the number of ones in the binary expansion of k . The finite product identity

$$P_n(x) := \prod_{j=0}^{n-1} (1 - e^{2\pi i 2^j x}) = \sum_{k=0}^{2^n-1} \tau_k e^{2\pi i k x} \quad (11)$$

follows by induction. Taking moduli yields

$$|P_n(x)| = 2^n \prod_{j=0}^{n-1} |s(T^j x)|. \quad (12)$$

Thus the same products governing the decay of $\widehat{\text{up}}$ are the normalized moduli of classical Thue–Morse trigonometric polynomials. This is the bridge to Gelfond exponents, Riesz products, and thermodynamic formalism; see [7, 13, 6].

3.5 A second exact scaling identity

The behavior just to the right of a large dyadic integer is exposed by a different exact identity.

Lemma 3.3 (Dyadic-boundary identity). *Let $N = 2^{n-1}$ and $0 < z < N$. Then*

$$|\Phi(N + z)| = \frac{|\Phi(\frac{1}{2} + z/(2N))|}{|\Phi(z/(2N))|} \left(\frac{z}{N+z}\right)^n |\Phi(z)|. \quad (13)$$

Proof. For $0 \leq h < n$, the number $N/2^h$ is an integer, so

$$\left| \sin(\pi(N+z)/2^h) \right| = \left| \sin(\pi z/2^h) \right|.$$

The quotient of the first n sinc factors is therefore $(z/(N+z))^n$ in absolute value. Reindexing the remaining tails gives

$$\prod_{h=n}^{\infty} \operatorname{sinc}\left(\frac{\pi(N+z)}{2^h}\right) = \Phi\left(\frac{1}{2} + \frac{z}{2N}\right), \quad \prod_{h=n}^{\infty} \operatorname{sinc}\left(\frac{\pi z}{2^h}\right) = \Phi\left(\frac{z}{2N}\right).$$

Combining the finite and infinite parts proves the identity. $\square$

4 Exactly one extremum per unit interval

The Weierstrass product gives a short complete proof of the observed one-extremum property.

Theorem 4.1 (Strict logarithmic concavity between zeros). *The function Φ is positive and strictly decreasing on $(0, 1)$. For every integer $m \geq 1$, $|\Phi|$ has exactly one critical point $\xi_m \in (m, m+1)$. It is the strict maximum of $|\Phi|$ on that interval. Moreover, for every integer $m \geq 0$,*

$$\operatorname{sgn} \Phi(x) = (-1)^{s_2(m)}, \quad m < x < m+1. \quad (14)$$

For $m \geq 1$ the signed extremum is therefore $\Phi(\xi_m) = (-1)^{s_2(m)} M_m$, where

$$M_m := \max_{m \leq x \leq m+1} |\Phi(x)|.$$

Proof. Away from the integer zeros, termwise logarithmic differentiation of (4) is locally uniform and gives

$$\frac{d}{dx} \log |\Phi(x)| = -2x \sum_{k=1}^{\infty} \frac{1 + \nu_2(k)}{k^2 - x^2}, \quad (15)$$

$$\frac{d^2}{dx^2} \log |\Phi(x)| = -2 \sum_{k=1}^{\infty} (1 + \nu_2(k)) \frac{k^2 + x^2}{(k^2 - x^2)^2} < 0. \quad (16)$$

For $0 < x < 1$, every denominator in (15) is positive, so $(\log \Phi)'(x) < 0$. This proves strict decrease on the first interval.

Now let $m \geq 1$. The logarithmic derivative is strictly decreasing on $(m, m+1)$. The pole contributed by $k = m$ forces it to $+\infty$ at the left endpoint, while the pole contributed by $k = m+1$ forces it to $-\infty$ at the right endpoint. It therefore has exactly one zero, and strict log-concavity makes that point the unique maximum of the modulus.

Starting from $\Phi(0) = 1$, crossing the zero at k changes the sign $1 + \nu_2(k)$ times. Therefore the sign on $(m, m+1)$ is

$$(-1)^{\sum_{k=1}^m (1 + \nu_2(k))}.$$

Legendre's identity $\nu_2(m!) = m - s_2(m)$ gives

$$\sum_{k=1}^m (1 + \nu_2(k)) = m + \nu_2(m!) = 2m - s_2(m),$$

whose parity is $s_2(m)$. This proves (14). $\square$

Remark 4.2. The critical points do *not* in general converge to the midpoints of their unit intervals. In [Section 12](#) we prove that, in every fixed interval offset from a large power of two, the critical point approaches the *right* endpoint at rate $1/n$.

5 Dyadic-shell means and the transfer operator

5.1 Normalized shell means

For $q \in (0, \infty)$ and a measurable H on the shell $D_n = [2^{n-1}, 2^n]$, define

$$\mathcal{M}_{q,n}(H) := \left(\frac{1}{2^{n-1}} \int_{2^{n-1}}^{2^n} |H(x)|^q dx \right)^{1/q} = \left(2 \int_{1/2}^1 |H(2^n y)|^q dy \right)^{1/q}. \quad (17)$$

The endpoint conventions are

$$\mathcal{M}_{0,n}(H) := \exp \left(\frac{1}{2^{n-1}} \int_{2^{n-1}}^{2^n} \log |H(x)| dx \right), \quad (18)$$

$$\mathcal{M}_{\infty,n}(H) := \sup_{2^{n-1} \leq x \leq 2^n} |H(x)|. \quad (19)$$

The logarithm in (18) is locally integrable despite the finitely many zeros.

After $y = (x + 1)/2$, the numerator in (10) becomes

$$\prod_{j=0}^{n-1} s(T^j x), \quad 0 \leq x \leq 1. \quad (20)$$

This product is generated by an explicit Perron–Frobenius operator.

5.2 The weighted doubling operator

For $q > 0$, define

$$(\mathcal{L}_q f)(x) := \frac{1}{2} \left[\sin^q \left(\frac{\pi x}{2} \right) f(x/2) + \cos^q \left(\frac{\pi x}{2} \right) f((x+1)/2) \right]. \quad (21)$$

The factor $1/2$ makes this the Lebesgue Perron operator with weight s^q . For every integrable f ,

$$\int_0^1 f(x) \prod_{j=0}^{n-1} s(T^j x)^q dx = \int_0^1 (\mathcal{L}_q^n f)(x) dx. \quad (22)$$

We use the following standard Ruelle–Perron–Frobenius fact. The weights in (21) are nonnegative rather than strictly positive at the two endpoints, but the two-branch system is primitive, so the usual cone and Lasota–Yorke proof applies without alteration; see [3, Chs. 2–3] and the singular-potential discussion in [5, 6].

Proposition 5.1 (Perron root and spectral decomposition). *For every $q > 0$, $\mathcal{L}_q$ has a simple leading eigenvalue $\rho_q > 0$, a strictly positive eigenfunction h_q , and a positive eigenmeasure ν_q , which may be normalized by*

$$\mathcal{L}_q h_q = \rho_q h_q, \quad \mathcal{L}_q^* \nu_q = \rho_q \nu_q, \quad \nu_q(h_q) = 1.$$

On a suitable Holder space $C^\alpha[0, 1]$, with $0 < \alpha \leq \min(1, q)$, there is a spectral gap and

$$\rho_q^{-n} \mathcal{L}_q^n f = \nu_q(f) h_q + \mathcal{O}(\vartheta_q^n \|f\|_{C^\alpha}), \quad 0 < \vartheta_q < 1. \quad (23)$$

The same conclusion applies to nonnegative endpoint-power functions encountered below by choosing a compatible Holder exponent.

5.3 The full finite- q decay spectrum

Set

$$\Lambda_q := \rho_q^{1/q}, \quad \kappa_q := \frac{\log \pi + a/2 - \log \Lambda_q}{a}, \quad q > 0. \quad (24)$$

Theorem 5.2 (Sharp finite- q shell asymptotics). *For every $q \in (0, \infty)$ there is a constant $A_q \in (0, \infty)$ such that*

$$\mathcal{M}_{q,n} \left(\frac{\Phi}{E_{\kappa_q}} \right) \rightarrow A_q \quad (n \rightarrow \infty). \quad (25)$$

More explicitly, with

$$\widetilde{W}_q(x) := W_{\kappa_q} \left(\frac{x+1}{2} \right)^q,$$

and the normalization in [Proposition 5.1](#),

$$A_q^q = \left(\int_0^1 h_q(x) dx \right) \nu_q(\widetilde{W}_q). \quad (26)$$

The number Λ_q is nondecreasing in q , and therefore κ_q is nonincreasing.

Proof. Raise (10) to the q th power, use (17), and substitute $x = 2y - 1$. This gives

$$\begin{aligned} \mathcal{M}_{q,n} \left(\frac{\Phi}{E_{\kappa}} \right)^q &= \exp(qn(\kappa a - \log \pi - a/2)) \\ &\times \int_0^1 W_{\kappa} \left(\frac{x+1}{2} \right)^q \prod_{j=0}^{n-1} s(T^j x)^q dx. \end{aligned}$$

By (22) and (23), the integral equals

$$\rho_q^n \left(\int_0^1 h_q \right) \nu_q(W_{\kappa}((\cdot + 1)/2)^q) (1 + o(1)).$$

The definition (24) is exactly the cancellation condition

$$q(\kappa_q a - \log \pi - a/2) + \log \rho_q = 0.$$

This proves (25) and (26).

For each n , the L^q norm of the product (20) on the probability space $[0, 1]$ is nondecreasing in q . Taking n th roots and then the limit proves that Λ_q is nondecreasing. Formula (24) then gives the opposite monotonicity of κ_q . $\square$

Remark 5.3 (No logarithmic-power correction). For a fixed $q \in (0, \infty)$, replacing E_{κ_q} by $E_{\kappa_q}(\log x)^p$ multiplies the normalized shell mean by $(na)^{-p}(1 + o(1))$. It therefore tends to 0 when $p > 0$ and to $+\infty$ when $p < 0$. At the sharp power, the required logarithmic exponent is $p = 0$.

6 Geometric-mean and typical decay

The $q = 0$ endpoint is especially simple because Lebesgue measure is invariant under the doubling map and

$$\int_0^1 \log \sin(\pi x) dx = -\log 2 = -a. \quad (27)$$

Theorem 6.1 (Exact shell geometric mean). *Define*

$$\kappa_0 := \frac{3}{2} + \frac{\log \pi}{a} = 3.151496129472319 \dots \quad (28)$$

Then

$$\mathcal{M}_{0,n} \left(\frac{\Phi}{E_{\kappa_0}} \right) = \exp \left(\int_0^1 \log W_{\kappa_0} \left(\frac{x+1}{2} \right) dx \right) \quad (29)$$

for every $n \geq 1$. In particular, the normalized shell geometric mean is not merely asymptotically constant; it is exactly independent of the shell.

Proof. Take logarithms in (10), substitute $x = 2y - 1$, and average. The n logarithmic sine terms each have integral $-a$ by invariance and (27). For (28),

$$\kappa_0 a - \log \pi - a/2 = a,$$

so the linear contribution na cancels the integral $-na$ exactly. $\square$

The same exponent governs almost every fixed dyadic phase.

Corollary 6.2 (Almost-everywhere dyadic-ray decay). *For Lebesgue-a.e. $y \in [1/2, 1]$,*

$$\log |\Phi(2^n y)| = -\frac{(\log(2^n y))^2}{2a} - \kappa_0 \log(2^n y) + o(n). \quad (30)$$

Equivalently,

$$|\Phi(2^n y)| = E_{\kappa_0}(2^n y) e^{o(n)}.$$

Proof. The function $\log s$ is integrable and the doubling map is ergodic. Birkhoff's theorem therefore gives

$$\frac{1}{n} \log B_n(y) = \frac{1}{n} \sum_{j=1}^n \log s(T^j y) \longrightarrow -a$$

for almost every y . Substitute this in (10) with $\kappa = \kappa_0$. $\square$

The $o(n)$ in (30) is genuinely dynamical; it need not converge to a constant. Thus even the “typical” statement is a logarithmic asymptotic, not a pointwise equivalent with one universal prefactor.

7 Central limit theorem and the corrected iterated-logarithm law

The almost-everywhere exponent records only the mean of the logarithmic cocycle. Its next order is probabilistic, but in this particular problem the variance and the martingale structure are completely explicit. Put

$$\psi(t) := \log |2 \sin(\pi t)|, \quad S_n(t) := \sum_{j=0}^{n-1} \psi(T^j t), \quad Tt = 2t \pmod{1}. \quad (31)$$

The exact centered logarithm in the alternative shell coordinates $x = 2^k(1+t)$ is

$$\log |\Phi(x)| = -\frac{(\log x)^2}{2a} - \kappa_0 \log x + \Psi(1+t) + S_{k+1}(t), \quad (32)$$

where the mantissa term is explicitly

$$\Psi(y) := \log W_{\kappa_0}(y/2), \quad 1 < y < 2. \quad (33)$$

It is bounded on every compact subinterval of $(1, 2)$. Thus all metric fluctuations are those of S_n .

7.1 Exact covariance and finite- n variance

The classical Fourier expansion

$$\psi(t) = - \sum_{m=1}^{\infty} \frac{\cos(2\pi mt)}{m} \quad \text{in } L^2(0, 1) \quad (34)$$

contains the whole calculation.

Theorem 7.1 (Exact variance). *The function ψ has mean zero and, for every $n \geq 1$,*

$$\int_0^1 S_n(t)^2 dt = \frac{\pi^2}{4}n - \frac{\pi^2}{3}(1 - 2^{-n}). \quad (35)$$

Consequently the asymptotic variance per dyadic level is

$$\sigma^2 = \lim_{n \rightarrow \infty} \frac{1}{n} \int_0^1 S_n^2 = \frac{\pi^2}{4}. \quad (36)$$

Proof. The mean is zero by the classical log-sine integral. Parseval and $\int_0^1 \cos^2(2\pi mt) dt = 1/2$ give

$$C_0 := \int_0^1 \psi(t)^2 dt = \frac{1}{2} \sum_{m \geq 1} \frac{1}{m^2} = \frac{\pi^2}{12}.$$

For $r \geq 1$, orthogonality leaves only frequencies $m = 2^r \ell$:

$$C_r := \int_0^1 \psi(t)\psi(T^r t) dt = \frac{1}{2} \sum_{\ell \geq 1} \frac{1}{2^r \ell} \frac{1}{\ell} = \frac{\pi^2}{12 \cdot 2^r}.$$

Stationarity now yields

$$\int S_n^2 = nC_0 + 2 \sum_{r=1}^{n-1} (n-r)C_r.$$

Since $\sum_{r=1}^{n-1} (n-r)2^{-r} = n - 2 + 2^{1-n}$, this is exactly (35). $\square$

7.2 A one-line Gordin decomposition

Let $\mathcal{P}$ be the unweighted Perron operator of the doubling map,

$$(\mathcal{P}f)(x) = \frac{1}{2} (f(x/2) + f((x+1)/2)).$$

From (34), $\mathcal{P}\psi = \psi/2$. Therefore

$$d := 2\psi - \psi \circ T = \log |\tan(\pi t)|, \quad \mathcal{P}d = 0, \quad (37)$$

and telescoping gives the exact identity

$$S_n = \sum_{j=0}^{n-1} d \circ T^j - \psi + \psi \circ T^n. \quad (38)$$

The sequence $(d \circ T^j)$ is a stationary ergodic reverse martingale-difference sequence for the decreasing filtration $(T^{-j}\mathcal{B})$. Moreover

$$\int_0^1 d(t)^2 dt = \frac{2}{\pi} \int_0^{\pi/2} \log^2(\tan u) du = \frac{\pi^2}{4}. \quad (39)$$

The final integral follows from the standard log-sine and log-cosine integrals; it also agrees with (36), as it must.

Theorem 7.2 (CLT and corrected LIL). *With respect to Lebesgue measure on $t \in (0, 1)$,*

$$\frac{S_n}{\sqrt{n}} \Rightarrow N\left(0, \frac{\pi^2}{4}\right). \quad (40)$$

For Lebesgue-a.e. t ,

$$\limsup_{n \rightarrow \infty} \frac{S_n(t)}{\sqrt{2n \log \log n}} = \frac{\pi}{2}, \quad \liminf_{n \rightarrow \infty} \frac{S_n(t)}{\sqrt{2n \log \log n}} = -\frac{\pi}{2}. \quad (41)$$

Equivalently, with denominator $\sqrt{n \log \log n}$, the two constants are $\pm\pi/\sqrt{2}$.

Proof. The tail estimate $\text{Leb}\{|d| > u\} = O(e^{-u})$ shows that $d \in L^p(0, 1)$ for every finite p . With $\mathcal{F}_j = T^{-j}\mathcal{B}$, the variables $d \circ T^j$ form a stationary reverse martingale-difference sequence. Moreover,

$$\frac{1}{n} \sum_{j=0}^{n-1} \mathbb{E}[(d \circ T^j)^2 \mid \mathcal{F}_{j+1}] = \frac{1}{n} \sum_{j=0}^{n-1} (\mathcal{P}d^2) \circ T^{j+1} \longrightarrow \int_0^1 d^2$$

almost surely by the ergodic theorem, and the conditional Lindeberg condition follows from $d \in L^2$. The stationary martingale CLT and the martingale LIL therefore apply to $\sum_{j < n} d \circ T^j$; see Stout [17] and Heyde–Scott [18]. The variance is (39). It remains only to remove the two coboundary terms in (38). The positive part of ψ is bounded by $\log 2$, while near the integers

$$\text{Leb}\{\psi < -u\} = O(e^{-u}).$$

Hence $\sum_n \text{Leb}\{|\psi \circ T^n| > \varepsilon \sqrt{n \log \log n}\} < \infty$. Borel–Cantelli makes the boundary terms $o(\sqrt{n \log \log n})$ almost surely. Replacing the threshold by $\varepsilon \sqrt{n}$ also makes them $o(\sqrt{n})$ in probability, which completes the CLT. $\square$

In terms of the Fourier transform, for a.e. fixed $y = 1 + t \in (1, 2)$ and $x_k = 2^k y$, (32) and (41) give

$$\limsup_{k \rightarrow \infty} \frac{\log |\Phi(x_k)| + (\log x_k)^2 / (2a) + \kappa_0 \log x_k - \Psi(y)}{\sqrt{2(k+1) \log \log(k+1)}} = \frac{\pi}{2}, \quad (42)$$

with the corresponding negative liminf. The multiplicative fluctuation is therefore

$$\exp\left(\pm \left(\frac{\pi}{\sqrt{2}} + o(1)\right) \sqrt{\frac{\log x}{\log 2} \log \log \log x}\right),$$

which is subpolynomial in x but larger than every fixed power of $\log x$.

7.3 The normalization error in the published theorem

Theorem 1.2 of Aistleitner–Hofer–Larcher [15] prints the upper and recurrent lower constants as π when the denominator is $\sqrt{n \log \log n}$. That value is incompatible with the exact variance (35). The paper's own variance calculation effectively uses $\int_0^1 \cos^2(2\pi mt) dt = 1$ rather than $1/2$, doubling σ^2 from $\pi^2/4$ to $\pi^2/2$. Restoring the missing factor gives $\sigma = \pi/2$ and therefore the constant $\sqrt{2}\sigma = \pi/\sqrt{2}$ in the $\sqrt{n \log \log n}$ normalization. This correction changes no exponent or qualitative conclusion in that paper, but it matters for the sharp constant.

8 The exact RMS power

For $q = 2$, the transfer operator simplifies dramatically:

$$\mathcal{L}_2 \mathbf{1} = \frac{1}{2} \left(\sin^2 \frac{\pi x}{2} + \cos^2 \frac{\pi x}{2} \right) = \frac{1}{2} \mathbf{1}. \quad (43)$$

Its operator norm on $C[0, 1]$ is at most $1/2$, so

$$\rho_2 = \frac{1}{2}, \quad \Lambda_2 = \frac{1}{\sqrt{2}}. \quad (44)$$

There is also a completely finite Fourier proof. Parseval applied to (11) gives

$$\int_0^1 |P_n(x)|^2 dx = 2^n,$$

while (12) gives

$$\int_0^1 \prod_{j=0}^{n-1} s(T^j x)^2 dx = 2^{-n}. \quad (45)$$

Corollary 8.1 (RMS decay). *The sharp root-mean-square exponent is*

$$\kappa_2 = \frac{\log(2\pi)}{\log 2} = 2.651496129472318 \dots \quad (46)$$

There is a constant $A_2 \in (0, \infty)$ such that

$$\mathcal{M}_{2,n} \left(\frac{\Phi}{E_{\kappa_2}} \right) \longrightarrow A_2. \quad (47)$$

No power of $\log x$ belongs in the leading RMS normalization.

Remark 8.2 (What the draft computed). The power $x^{-\log(2\pi)/\log 2}$ in [12] is precisely $x^{-\kappa_2}$. Thus the draft identified the quadratic-mean scale, albeit without specifying that norm. Its claimed pointwise equivalent and its proposed logarithmic correction do not follow and are incompatible with the exact shell identity (45), the integer zeros, and the exceptional extrema proved later.

9 Arithmetic-mean decay and the Cesaro question

9.1 The Perron root ρ_1

For $q = 1$,

$$(\mathcal{L}_1 f)(x) = \frac{1}{2} \left[\sin \frac{\pi x}{2} f(x/2) + \cos \frac{\pi x}{2} f((x+1)/2) \right]. \quad (48)$$

There appears to be no elementary closed form for its Perron root. It is nevertheless an exact, canonical constant, and it is easy to compute stably:

$$\rho_1 = 0.661322602060565 \dots \quad (49)$$

The corresponding power is

$$\kappa_1 = \frac{\log \pi + a/2 - \log \rho_1}{a} = 2.748070014871334 \dots \quad (50)$$

The decimal in (49) is numerical, but the following elementary Collatz–Wielandt certificate provides a rigorous enclosure.

Proposition 9.1 (A rational enclosure for ρ_1). *One has*

$$0.66126807 < \rho_1 < 0.66134891, \quad (51)$$

and consequently

$$2.74801262 < \kappa_1 < 2.74818899. \quad (52)$$

Proof. For a positive continuous test function f , the Collatz–Wielandt inequalities give

$$\inf_{x \in [0,1]} \frac{\mathcal{L}_1 f(x)}{f(x)} \leq \rho_1 \leq \sup_{x \in [0,1]} \frac{\mathcal{L}_1 f(x)}{f(x)}.$$

Take

$$f(x) = F(\sin \pi x), \quad F(u) = 1 + \frac{376189}{10^6}u - \frac{69093}{10^6}u^2 + \frac{15483}{10^6}u^3.$$

This function is positive on $[0, 1]$. Put $t = \tan(\pi x/4) \in [0, 1]$, $u = 2t/(1+t^2)$, and $v = (1-t^2)/(1+t^2)$. Then

$$\frac{\mathcal{L}_1 f(x)}{f(x)} = \frac{uF(u) + vF(v)}{2F(2uv)},$$

a rational function of t . Exact Sturm-sequence checks show that the numerators of its differences from the two rational endpoints in (51) are positive and have no zeros on $[0, 1]$. A compact exact-arithmetic verification is included in [Section A](#). $\square$

Remark 9.2 (Published enclosure). Aistleitner–Hofer–Larcher prove the sharper rigorous interval

$$0.66130 < \rho_1 < 0.66135, \quad (53)$$

by the Fouvry–Mauduit transfer recursion. It implies $2.74801024 < \kappa_1 < 2.74811933$. The wider interval above is retained because its certificate is short, exact, and executable from the appendix alone.

9.2 Sharp dyadic-shell mean

By [Theorem 5.2](#), there is $A_1 > 0$ such that

$$2 \int_{1/2}^1 \frac{|\Phi(2^n y)|}{E_{\kappa_1}(2^n y)} dy \rightarrow A_1. \quad (54)$$

Therefore the choice

$$g_1(x) := A_1 E_{\kappa_1}(x) \quad (55)$$

makes the mean of $|\Phi|/g_1$ on each dyadic shell tend to one. Chebyshev collocation of the transfer operator, followed by the spectral formula (26), gives

$$A_1 = 0.091266124131588 \dots \quad (56)$$

The decimal is numerical; the exact definition is the positive spectral expression (26). This gives the precise answer to the natural shell version of the question in [10].

Proposition 9.3 (Regularity of the arithmetic-mean gauge). *The function $g_1(x) = A_1 E_{\kappa_1}(x)$ is positive and real-analytic on $(0, \infty)$ and strictly decreasing for $x \geq 1$. For every fixed integer $m \geq 0$ there is X_m such that*

$$(-1)^m g_1^{(m)}(x) > 0, \quad x \geq X_m. \quad (57)$$

Thus it has every one of the eventual regularity properties requested in the original question, although the threshold may depend on the derivative order.

Proof. Put $u = \log x$ and $G(u) = g_1(e^u)$. Then

$$G(u) = A_1 \exp\left(-\frac{u^2}{2a} - \kappa_1 u\right), \quad x^m \frac{d^m}{dx^m} = D_u(D_u - 1) \cdots (D_u - m + 1).$$

Writing $p(u) = u/a + \kappa_1$, repeated differentiation gives

$$(-1)^m \frac{x^m g_1^{(m)}(x)}{g_1(x)} = p(u)^m + O_m(p(u)^{m-1}) \quad (u \rightarrow \infty).$$

The right-hand side is positive for all sufficiently large u . The case $m = 1$ also follows directly from $g_1'(x)/g_1(x) = -(\log x/a + \kappa_1)/x < 0$ for $x \geq 1$. $\square$

It also answers the weaker full-ray boundedness question.

Corollary 9.4 (Two-sided Cesaro boundedness). *For every fixed $t_0 > 0$, there are constants $0 < c < C < \infty$ such that, for all sufficiently large t ,*

$$c < \frac{1}{t - t_0} \int_{t_0}^t \frac{|\Phi(x)|}{E_{\kappa_1}(x)} dx < C. \quad (58)$$

After multiplying E_{κ_1} by A_1 , the same average tends to one along the dyadic endpoints $t = 2^n$.

Proof. Equation (54) says that the integral over the shell $[2^{n-1}, 2^n]$ is asymptotic to $A_1 2^{n-1}$. Summing the geometric sequence of completed shells proves the dyadic-endpoint limit. If t lies inside the next shell, the completed shells already contribute a positive constant times t , while the entire current shell contributes at most another constant times t . This gives (58). $\square$

9.3 Why the ordinary Cesaro limit has a dyadic phase

The full limit in the stronger condition of [10] does not follow from the shell limit. In fact, the natural normalization (55) has a nonconstant phase profile. This can be stated exactly in terms of the eigenmeasure of $\mathcal{L}_1$.

Normalize h_1, ν_1 as in Proposition 5.1, and put

$$f_1(x) := W_{\kappa_1}\left(\frac{x+1}{2}\right), \quad \mathcal{F}(z) := \frac{\nu_1(f_1 \mathbf{1}_{[0,z]})}{\nu_1(f_1)}, \quad 0 \leq z \leq 1.$$

The measure $f_1 d\nu_1$ is non-atomic, so $\mathcal{F}$ is continuous. Here is a short verification. The endpoint atoms of ν_1 vanish because the corresponding transfer weights vanish. For $z \in (0, 1) \setminus \{1/2\}$, the eigenmeasure equation gives

$$\rho_1 \nu_1(\{z\}) = \frac{1}{2} s(z) \nu_1(\{Tz\}).$$

(The possible atom at $1/2$ is fed only by the already vanishing endpoint atoms.) Hence any remaining atom would generate forward atoms whose masses grow by at least the factor $2\rho_1 > 1$ at every step. A periodic orbit gives an immediate contradiction, and a nonperiodic orbit would produce masses exceeding the finite total mass. Thus ν_1 , and therefore $f_1 d\nu_1$, has no atoms.

Theorem 9.5 (Dyadic phase limit). *For every $y \in [1/2, 1]$ and every fixed $t_0 > 0$,*

$$\lim_{n \rightarrow \infty} \frac{1}{2^n y - t_0} \int_{t_0}^{2^n y} \frac{|\Phi(x)|}{A_1 E_{\kappa_1}(x)} dx = \mathcal{P}(y) := \frac{1 + \mathcal{F}(2y - 1)}{2y}. \quad (59)$$

The continuous function $\mathcal{P}$ satisfies $\mathcal{P}(1/2) = \mathcal{P}(1) = 1$ but is not identically one. Consequently the ordinary Cesaro limit for the natural elementary scale $A_1 E_{\kappa_1}$ does not exist.

Proof. The spectral decomposition gives, for $z \in [0, 1]$,

$$\rho_1^{-n} \int_0^z f_1(x) \prod_{j=0}^{n-1} s(T^j x) dx \longrightarrow \left(\int_0^1 h_1 \right) \nu_1(f_1 \mathbf{1}_{[0,z]}).$$

Approximation from above and below by Holder functions justifies the indicator; the leading conformal measure has no atoms. Returning to the shell variable, the partial integral from 2^{n-1} to $2^n y$ is asymptotic to

$$2^{n-1} A_1 \mathcal{F}(2y - 1).$$

The sum of all preceding complete shells is asymptotic to $2^{n-1} A_1$. Division by $2^n y A_1$ gives (59).

It remains to prove that $\mathcal{F}$ is not the identity distribution. We first note that ν_1 is singular with respect to Lebesgue measure. Otherwise its nonzero absolutely continuous part would yield an L^1 density $r \geq 0$ satisfying

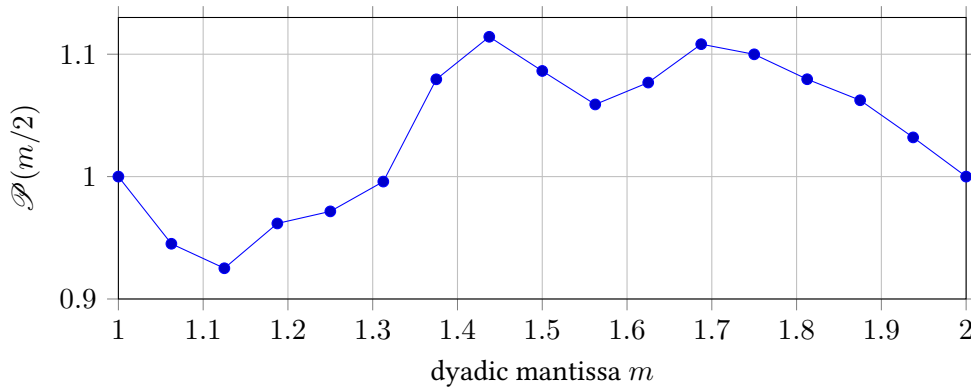
$$\rho_1 r(x) = s(x) r(Tx) \quad \text{for a.e. } x. \quad (60)$$

On the positive invariant support of r , iteration gives

$$r(T^n x) = \frac{\rho_1^n r(x)}{\prod_{j=0}^{n-1} s(T^j x)}.$$

For Lebesgue-a.e. x , Birkhoff's theorem makes the logarithmic growth rate of the right side $\log(2\rho_1) > 0$. On the other hand, for any $r \in L^1$ and any $\epsilon > 0$, Markov's inequality and Borel–Cantelli imply $r(T^n x) \leq e^{\epsilon n}$ eventually for almost every x . Choosing $0 < \epsilon < \log(2\rho_1)$ is a contradiction. Thus ν_1 is singular, and multiplying it by the continuous function f_1 leaves it singular. Hence its normalized distribution $\mathcal{F}$ cannot equal z , the distribution function of Lebesgue measure. But $\mathcal{P} \equiv 1$ would force $1 + \mathcal{F}(2y - 1) = 2y$, or $\mathcal{F}(z) = z$. Therefore $\mathcal{P}$ is nonconstant. $\square$

For orientation, the following finite-level computation displays the nonconstant profile in terms of the shell mantissa $m = 2y \in [1, 2]$. It uses the normalized density $f_1(z) \prod_{j < 18} s(T^j z)$ on a dyadic midpoint grid; the theorem, not the plot, supplies the limiting statement.



Remark 9.6. A slowly varying correction, including every power of $\log x$, cannot remove this phase. For a slowly varying L , the ratio $L(2^n y)/L(2^n)$ tends uniformly to one for $y \in [1/2, 1]$, so the profile (59) survives. An exact all-phase normalizer would have to encode the singular conformal distribution, not merely add a smooth logarithmic factor.

9.4 How far the unrestricted Cesaro obstruction extends

The nonconstant profile in Theorem 9.5 is not an artifact of having omitted a power of $\log x$. It survives every gauge whose relative shape settles down inside successive dyadic shells.

Theorem 9.7 (Impossibility in the stabilizing class). *Let $g : (X_0, \infty) \rightarrow (0, \infty)$ be a gauge. Suppose there are constants $c_k > 0$ and a continuous $u : [0, 1] \rightarrow (0, \infty)$ such that*

$$\frac{g(2^k(1+z))}{c_k E_{\kappa_1}(2^k(1+z))} \longrightarrow u(z) \quad \text{uniformly for } 0 \leq z \leq 1. \quad (61)$$

Then the unrestricted limit

$$\frac{1}{t - X_0} \int_{X_0}^t \frac{|\Phi(x)|}{g(x)} dx \longrightarrow 1 \quad (62)$$

cannot hold.

Proof. Assume (62). If $F(t)$ denotes the numerator integral, then for every fixed $0 \leq z_1 < z_2 \leq 1$,

$$\frac{F(2^k(1+z_2)) - F(2^k(1+z_1))}{2^k} \longrightarrow z_2 - z_1. \quad (63)$$

The exact factorization in the coordinates $x = 2^k(1+z)$ gives

$$\frac{|\Phi(2^k(1+z))|}{E_{\kappa_1}(2^k(1+z))} = \rho_1^{-(k+1)} W(z) \prod_{j=0}^k |\sin(\pi 2^j z)|,$$

where W is continuous and positive on $[0, 1)$. Combining this identity with (61) and the Ruelle–Perron–Frobenius limit shows that every subsequential shell limit in (63) is a constant multiple of

$$\nu_1 \left(\frac{W}{u} \mathbf{1}_{[z_1, z_2]} \right),$$

where ν_1 is the eigenmeasure of $\mathcal{L}_1^*$. The full-shell case forces the multiplicative constant to be finite and nonzero. Equation (63) would therefore make the measure $(W/u)\nu_1$ proportional to Lebesgue measure. Since W/u is continuous and positive on $[0, 1)$ and ν_1 has no endpoint atoms, this would imply $\nu_1 \ll \text{Leb}$.

That is impossible. If $d\nu_1 = r dx$ had a nonzero absolutely continuous part, the eigenmeasure equation would give

$$|\sin(\pi z)| r(Tz) = \rho_1 r(z) \quad \text{a.e.} \quad (64)$$

On the positive support of r , iteration and Birkhoff's theorem yield

$$r(T^n z) = r(z) \frac{\rho_1^n}{\prod_{j=0}^{n-1} |\sin(\pi T^j z)|} = r(z) \exp(n(\log(2\rho_1) + o(1))).$$

Here $2\rho_1 > 1$; for example, Cauchy–Schwarz together with the exact L^2 identity and the Gelfond upper bound gives $\rho_1 \geq 1/\sqrt{3} > 1/2$. On the other hand, invariance of Lebesgue measure, Markov's inequality, and Borel–Cantelli imply $r(T^n z) \leq e^{\varepsilon n}$ eventually for a.e. z , for every $\varepsilon > 0$. Choosing $\varepsilon < \log(2\rho_1)$ is a contradiction. Thus ν_1 is singular and (62) cannot hold. $\square$

The hypothesis covers all gauges of the form $E_{\kappa_1}(x)L(x)p(\{\log_2 x\})$, where L is regularly varying (slow variation is the most relevant case) and p is positive, continuous, and periodic. It also covers the usual log-exp-algebraic or Hardy-field candidates of the correct rough size whenever their relative logarithmic derivative has a finite limit, because the mean-value theorem then gives a uniform limiting shell shape. It does not, by itself, rule out a deliberately scale-dependent analytic gauge whose within-shell shape keeps changing and attempts to follow finer and finer dyadic cylinders. None of the source materials proves such a fully unrestricted nonexistence theorem.

9.5 The multiplicatively natural mean

The endpoint obstruction disappears if additive Cesaro averaging is replaced by logarithmic averaging. Each dyadic shell then has equal weight, and the final incomplete shell becomes negligible.

Theorem 9.8 (Unrestricted logarithmic mean). *There is a constant $A_1^{\log} > 0$ such that, for every fixed $t_0 > 1$,*

$$\frac{1}{\log(t/t_0)} \int_{t_0}^t \frac{|\Phi(x)|}{E_{\kappa_1}(x)} \frac{dx}{x} \longrightarrow A_1^{\log} \quad (t \rightarrow \infty) \quad (65)$$

without any restriction on the endpoint. In the normalization of [Proposition 5.1](#), if $f_1(x) = W_{\kappa_1}((x+1)/2)$, then

$$A_1^{\log} = \frac{1}{\log 2} \left(\int_0^1 h_1(x) dx \right) \nu_1 \left(\frac{f_1}{1+\cdot} \right). \quad (66)$$

Numerically,

$$A_1^{\log} = 0.09386063575678 \dots \quad (67)$$

Proof. Let

$$\beta_n = \int_{2^{n-1}}^{2^n} \frac{|\Phi(x)|}{E_{\kappa_1}(x)} \frac{dx}{x}.$$

The master factorization and $x = 2^n y$, followed by $z = 2y - 1$, give

$$\beta_n = \rho_1^{-n} \int_0^1 \frac{f_1(z)}{1+z} \prod_{j=0}^{n-1} |\sin(\pi T^j z)| dz.$$

The spectral limit in [Proposition 5.1](#) yields $\beta_n \rightarrow \beta_\infty$, where $\beta_\infty = (\int h_1) \nu_1(f_1/(1+\cdot))$. If $2^N \leq t < 2^{N+1}$, the logarithmic integral equals $\sum_{n < N} \beta_n + O(1)$, while $\log(t/t_0) = N \log 2 + O(1)$. Ordinary Cesaro convergence of the sequence (β_n) proves (65) and (66). $\square$

10 The sharp uniform envelope

The L^∞ numerator rate can be proved by a sharp elementary inequality. Put

$$\lambda := \frac{\sqrt{3}}{2}. \quad (68)$$

Lemma 10.1 (Sharp two-step logarithmic inequality). *For every $x \in \mathbb{R}$,*

$$\frac{2}{3} \log s(x) + \frac{1}{3} \log s(Tx) \leq \log \lambda, \quad (69)$$

with the usual interpretation at zeros. Equality holds precisely when $x \equiv 1/3$ or $2/3 \pmod{1}$.

Proof. Cubing the desired inequality gives

$$s(x)^2 s(Tx) \leq \lambda^3.$$

With $\theta = \pi x$ and $u = \sin^2 \theta$,

$$s(x)^2 s(Tx) = 2 |\sin^3 \theta \cos \theta|,$$

whose square is $4u^3(1-u)$. The maximum of $u^3(1-u)$ on $[0, 1]$ is $27/256$, attained at $u = 3/4$. Therefore the maximum of the original expression is $3\sqrt{3}/8 = \lambda^3$, with equality at $x \equiv 1/3, 2/3$. $\square$

Proposition 10.2 (Uniform lacunary-product bound). *For every $n \geq 1$,*

$$\lambda^n \leq \sup_{x \in [0,1]} \prod_{j=0}^{n-1} s(T^j x) \leq \lambda^{n-1}. \quad (70)$$

Consequently

$$\lim_{n \rightarrow \infty} \left\| \prod_{j=0}^{n-1} s \circ T^j \right\|_{\infty}^{1/n} = \lambda = \frac{\sqrt{3}}{2}. \quad (71)$$

Proof. At $x = 1/3$ the orbit alternates between $1/3$ and $2/3$, and every factor equals λ , proving the lower bound. For the upper bound, sum (69) for $x, Tx, \dots, T^{n-2}x$. If $\phi = \log s$, the result is

$$3 \sum_{j=0}^{n-2} \phi(T^j x) - \phi(x) + \phi(T^{n-1}x) \leq 3(n-1) \log \lambda.$$

After adding the final factor,

$$\sum_{j=0}^{n-1} \phi(T^j x) \leq (n-1) \log \lambda + \frac{1}{3} \phi(x) + \frac{2}{3} \phi(T^{n-1}x) \leq (n-1) \log \lambda,$$

because $\phi \leq 0$. Exponentiation gives the upper bound. $\square$

This is the classical Gelfond exponent for the Thue–Morse polynomial: by (12), the unnormalized polynomial grows like $3^{n/2}$ in supremum norm. The ergodic-optimization formulation and its generalizations are developed in [6].

Theorem 10.3 (Sharp global upper envelope). *Define*

$$\kappa_{\infty} := \frac{\log \pi + a/2 - \log(\sqrt{3}/2)}{a} = \frac{3}{2} + \frac{\log(\pi/\sqrt{3})}{\log 2} = 2.359014879111741 \dots \quad (72)$$

Then there is a constant $C < \infty$ such that

$$|\Phi(x)| \leq C E_{\kappa_{\infty}}(x), \quad x \geq 1. \quad (73)$$

This exponent is sharp. In fact, for every $n \geq 0$,

$$|\Phi(2^n \cdot 2/3)| = C_* E_{\kappa_{\infty}}(2^n \cdot 2/3), \quad (74)$$

where

$$C_* = W_{\kappa_{\infty}}(2/3) = 0.139129774734829 \dots > 0. \quad (75)$$

Proof. In (10), the definition of κ_{∞} makes

$$\exp(n(\kappa_{\infty}a - \log \pi - a/2)) = \lambda^{-n}.$$

After the substitution $x = Ty$, the product is of the form bounded in (70); therefore

$$\frac{|\Phi(2^n y)|}{E_{\kappa_{\infty}}(2^n y)} \leq \lambda^{-1} \max_{1/2 \leq y \leq 1} W_{\kappa_{\infty}}(y).$$

This proves the uniform upper bound. At $y = 2/3$, the orbit alternates between $2/3$ and $1/3$, so $B_n(2/3) = \lambda^n$ exactly. Equation (10) then reduces to (74). $\square$

Corollary 10.4 (Universal logarithmic order). *With the convention $\log 0 = -\infty$,*

$$\limsup_{x \rightarrow \infty} \frac{\log |\Phi(x)|}{(\log x)^2} = -\frac{1}{2 \log 2}, \quad \liminf_{x \rightarrow \infty} \frac{\log |\Phi(x)|}{(\log x)^2} = -\infty. \quad (76)$$

Equivalently, the coefficient $1/(2 \log 2)$ is the sharp global log-square order, but no positive two-sided pointwise equivalent exists.

Proof. The upper bound in the limsup follows from (73); the exact period-two ray (74) gives the reverse inequality. The liminf is $-\infty$ because Φ vanishes at every positive integer (and also because one may approach those zeros arbitrarily closely). $\square$

Corollary 10.5 (Sharp smoothness class). *For every $M > 0$, $\Phi(x) = \mathcal{O}(x^{-M})$ as $x \rightarrow \infty$. On the other hand, for every $c > 0$, $e^{cx} |\Phi(x)|$ is unbounded. Thus the transform decays faster than every power but not at any fixed exponential rate.*

Proof. The first statement follows from (73), because the negative quadratic in $\log x$ dominates $-M \log x$. The exact peak ray (74) decays only like $\exp(-\mathcal{O}((\log x)^2))$, which is much larger than e^{-cx} along that subsequence. $\square$

11 The raw dyadic-annulus maximum and its Lambert boundary layer

The sharp pointwise envelope of Theorem 10.3 is evaluated at the actual argument x . A different question is to compare every value in the shell $[2^k, 2^{k+1}]$ with the envelope at its *left endpoint*. The maximizer then moves toward that endpoint and pays a second, slower log-normal cost. This is the main result of Wave 1, Document 2 [20]; we reproduce the proof because it is the most delicate asymptotic in the corpus.

Put

$$\mathfrak{M}_k := \max_{2^k \leq x \leq 2^{k+1}} |\Phi(x)|, \quad n := k + 1, \quad \beta := \log \frac{\sqrt{3}}{2} < 0. \quad (77)$$

Also define the tail factor

$$\mathcal{H}(y) := \prod_{m=1}^{\infty} \left| \operatorname{sinc} \left(\frac{\pi y}{2^m} \right) \right|, \quad 1 \leq y \leq 2.$$

11.1 Exact reduction and transition coordinates

Writing $x = 2^k(1 + \theta)$, $0 \leq \theta \leq 1$, gives

$$|\Phi(2^k(1 + \theta))| = \frac{2^{-k(k+1)/2}}{\pi^n} \mathcal{H}(1 + \theta)(1 + \theta)^{-n} Q_n(\theta), \quad (78)$$

where

$$Q_n(\theta) = \prod_{j=0}^{n-1} |\sin(\pi 2^j \theta)|.$$

Hence

$$\mathfrak{M}_k = \pi^{-n} 2^{-k(k+1)/2} A_n, \quad A_n := \sup_{0 \leq \theta \leq 1} \mathcal{H}(1 + \theta)(1 + \theta)^{-n} Q_n(\theta). \quad (79)$$

The weight $(1 + \theta)^{-n}$ pushes the maximum toward 0, where Q_n vanishes. For $0 < \theta \leq 1/2$, write uniquely up to endpoints

$$\theta = 2^{-r} s, \quad r \in \mathbb{Z}_{\geq 0}, \quad s \in I := [1/4, 1/2]. \quad (80)$$

If $r < n$, then

$$Q_n(2^{-r}s) = R_r(s)Q_{n-r}(s), \quad R_r(s) = \prod_{h=1}^r \sin\left(\frac{\pi s}{2^h}\right). \quad (81)$$

Uniformly for $s \in I$,

$$\log R_r(s) = -\frac{a}{2}r^2 + r\left(\log(\pi s) - \frac{a}{2}\right) + O(1). \quad (82)$$

The remaining tail satisfies $Q_m(s) \leq Ce^{m\beta}$ by [Proposition 10.2](#).

For a matching lower bound one needs near-maximizing tails at essentially arbitrary s .

Lemma 11.1 (Dense preimages of the maximizing cycle). *There is $C > 0$ such that for every $s_0 \in I$ and $q \geq 1$ one can choose $s_q \in I$ with*

$$|s_q - s_0| \leq 2^{-q}, \quad T^q s_q \in \{1/3, 2/3\}, \quad (83)$$

and, for every $m \geq q$,

$$Q_m(s_q) \geq \exp(m\beta - Cq^2). \quad (84)$$

Proof. The q -fold preimages $(j + 1/3)/2^q$ have spacing 2^{-q} ; together with the preimages of $2/3$ they approximate every point of I as in (83). Before time q , every orbit point is a nonintegral rational with reduced denominator at most $3 \cdot 2^{q-j}$. Therefore

$$|\sin(\pi T^j s_q)| \geq 2 \operatorname{dist}(T^j s_q, \mathbb{Z}) \geq \frac{2}{3 \cdot 2^{q-j}}.$$

The first q factors lose at most $\exp(O(q^2))$. Thereafter the orbit alternates between $1/3$ and $2/3$, and every factor equals $\sqrt{3}/2 = e^\beta$. $\square$

Define the finite-dimensional objective

$$\mathcal{F}_n(r, s) = -\frac{a}{2}r^2 + r\left(\log(\pi s) - \frac{a}{2} - \beta\right) - n \log(1 + s2^{-r}). \quad (85)$$

Lemma 11.2 (Reduction to the boundary-layer objective). *As $n \rightarrow \infty$,*

$$\log A_n = n\beta + \sup_{\substack{r \in \mathbb{Z}_{\geq 0} \\ s \in I}} \mathcal{F}_n(r, s) + O((\log \log n)^2). \quad (86)$$

Proof. The upper bound follows from (82), the uniform Gelfond bound, and the fact that $\theta \geq 1/2$ or $r \geq n$ is exponentially inferior to the trial point $r = \lfloor \log_2 n \rfloor$, $s = 1/3$. At every near-maximizer one obtains $r = O(\log n)$ and $ns2^{-r} = O((\log n)^2)$. For the lower bound, approximate its s by s_q from [Lemma 11.1](#) with $q = \lceil 4 \log_2 \log(n + e^e) \rceil$. The derivative of $\mathcal{F}_n$ in s is $O((\log n)^2)$ in the localized region, so the displacement 2^{-q} costs $O(1)$; the tail costs only $O(q^2) = O((\log \log n)^2)$. The bounded factors $\mathcal{H}$ and the finite sinc tail contribute $O(1)$. $\square$

11.2 Lambert optimization

Set

$$C_0 := \log \pi - \frac{a}{2} - \beta = \log\left(\pi \sqrt{\frac{2}{3}}\right), \quad c := ae^{-C_0} = \frac{a}{\pi} \sqrt{\frac{3}{2}}. \quad (87)$$

At the localized maximum, replacing $-n \log(1 + s2^{-r})$ by $-ns2^{-r}$ changes the supremum by $o(1)$. Put $A = C_0 + \log s$ and $z = ar - A$. Then

$$-\frac{a}{2}r^2 + rA - ns2^{-r} = -\frac{z^2}{2a} - ne^{-C_0}e^{-z} + \frac{A^2}{2a}. \quad (88)$$

As s traverses $[1/4, 1/2]$, the interval of possible A has length exactly a . Successive integer values of r therefore make the corresponding z -intervals tile the real line. Since $A^2/(2a) = O(1)$, the supremum is, up to $O(1)$, the real maximum of

$$\phi_n(z) = -\frac{z^2}{2a} - ne^{-C_0}e^{-z}.$$

Its critical equation is $ze^z = cn$, so the maximizer is $w_n = W(cn)$ and

$$\max \phi_n = -\frac{w_n^2 + 2w_n}{2a}. \quad (89)$$

Theorem 11.3 (Sharp dyadic-annulus maximum). *Let $w_k = W(c(k+1))$, with c from (87). Then*

$$\log \mathfrak{M}_k = -\frac{a}{2}k(k+1) - (k+1)\log \pi + (k+1)\beta - \frac{w_k^2 + 2w_k}{2a} + O((\log \log k)^2). \quad (90)$$

Equivalently,

$$\log \mathfrak{M}_k = -\frac{a}{2}k^2 - a\kappa_\infty k - \frac{W(c(k+1))^2 + 2W(c(k+1))}{2a} + O((\log \log k)^2). \quad (91)$$

Any maximizing phase θ_k satisfies

$$\theta_k = \frac{W(c(k+1))}{a(k+1)}e^{o(1)}. \quad (92)$$

Proof. Combine (79), Lemma 11.2, and (89). The compact form uses $a\kappa_\infty = a/2 + \log \pi - \beta$. Strict concavity of ϕ_n near w_n and the $O((\log \log n)^2)$ error localize z to $w_n + o(1)$. Since $\theta = s2^{-r} = e^{-C_0-z}$ and $ne^{-C_0-w_n} = w_n/a$, this gives (92). $\square$

Let $R = 2^k$, $L = \log R$, and $d = \pi^{-1}\sqrt{3/2}$. Since $c(k+1) = d(L+a)$, (91) becomes

$$\log \max_{R \leq x \leq 2R} |\Phi(x)| = -\frac{L^2}{2a} - \kappa_\infty L - \frac{W(d(L+a))^2 + 2W(d(L+a))}{2a} + O((\log \log \log R)^2). \quad (93)$$

The extra term begins

$$-\frac{(\log \log R)^2}{2a} + \frac{\log \log R \log \log \log R}{a} + O(\log \log R). \quad (94)$$

Thus the annular maximum is smaller than the left-endpoint envelope $E_{\kappa_\infty}(R)$ by a second log-normal factor, even though the pointwise envelope itself is attained exactly on the fixed ray $x = 2^n(2/3)$. There is no contradiction: the fixed ray is a constant fraction into its shell, whereas the raw annular maximizer migrates toward the left boundary according to (92).

12 Exceptional extrema near dyadic boundaries

The sharp envelope in Theorem 10.3 describes the largest possible values in a shell, but it does not make every local extremum comparable to one constant. The deepest and most explicit obstruction occurs immediately after powers of two.

Theorem 12.1 (Dyadic valleys: exact extremal asymptotic). *Fix an integer $r \geq 0$, and put*

$$b := r + 1, \quad d := 1 + \nu_2(b), \quad N := 2^{n-1}.$$

Let ξ_{N+r} be the unique critical point in $(N+r, N+r+1)$ and $M_{N+r} = |\Phi(\xi_{N+r})|$. Then, as $n \rightarrow \infty$,

$$\xi_{N+r} = N + b - \frac{db}{n} + o(n^{-1}), \quad (95)$$

$$M_{N+r} \sim \left| \Phi(1/2) \Phi(b/2^d) \right| \left(\frac{b}{N} \right)^n \left(\frac{d}{en} \right)^d. \quad (96)$$

In particular,

$$\log M_{N+r} = -\frac{(\log N)^2}{\log 2} + \mathcal{O}(\log N). \quad (97)$$

The leading quadratic coefficient in (97) is twice the coefficient in E_κ .

Proof. Write $x = N + z$, $r < z < b$, in the exact boundary identity (13). Uniformly for $z \in [r, b]$,

$$\frac{|\Phi(\frac{1}{2} + z/(2N))|}{|\Phi(z/(2N))|} \rightarrow |\Phi(1/2)|, \quad \left(1 + \frac{z}{N}\right)^{-n} \rightarrow 1, \quad (98)$$

because $n/N \rightarrow 0$. Hence

$$M_{N+r} = |\Phi(1/2)| N^{-n} \max_{r \leq z \leq b} (z^n |\Phi(z)|) (1 + o(1)). \quad (99)$$

By (6), as $z \uparrow b$,

$$|\Phi(z)| = D_b (b - z)^d (1 + \mathcal{O}(b - z)), \quad D_b = b^{-d} \left| \Phi(b/2^d) \right|.$$

Any fixed region $z \leq b - \delta$ is exponentially negligible relative to b^n , so the maximum lies in a shrinking neighborhood of b . Set $u = n(b - z)/b$. Uniformly for bounded u ,

$$z^n |\Phi(z)| = D_b b^{n+d} n^{-d} u^d \left(1 - \frac{u}{n}\right)^n (1 + o(1)) \rightarrow D_b b^{n+d} n^{-d} u^d e^{-u}.$$

The function $u^d e^{-u}$ has its unique maximum at $u = d$, with value $d^d e^{-d}$. Standard localization of the maximum therefore gives

$$b - z_{n,r} \sim \frac{db}{n}$$

and

$$\max z^n |\Phi(z)| \sim D_b b^{n+d} n^{-d} d^d e^{-d}.$$

Insert $D_b = b^{-d} |\Phi(b/2^d)|$ into (99) to obtain (96); the location statement is (95). Finally, $-n \log N = -(\log N)^2/a + \mathcal{O}(\log N)$, proving (97). $\square$

For example, in the first interval after $N = 2^{n-1}$, $r = 0$, $b = d = 1$, and

$$M_N \sim \frac{\Phi(1/2)^2}{en} N^{-n}, \quad \xi_N = N + 1 - \frac{1}{n} + o(n^{-1}). \quad (100)$$

Numerically, $\Phi(1/2) = 0.553771275888810 \dots$

12.1 Consequences for the sequence of extrema

Corollary 12.2 (The upper envelope is sharp but not two-sided). For $M_m = \max_{[m, m+1]} |\Phi|$,

$$0 < \limsup_{m \rightarrow \infty} \frac{M_m}{E_{\kappa\infty}(m)} < \infty, \quad \liminf_{m \rightarrow \infty} \frac{M_m}{E_{\kappa\infty}(m)} = 0. \quad (101)$$

Therefore no constant multiple of the sharp uniform envelope makes all local extrema approach one nonzero amplitude.

Proof. The upper bound follows from (73). Let $x_n = 2^n(2/3)$ and $m_n = \lfloor x_n \rfloor$. The local maximum in the containing unit interval is at least $|\Phi(x_n)|$, and (74) gives a positive lower bound after division by $E_{\kappa_\infty}(m_n)$; changing the argument by less than one changes E_{κ_∞} by a factor tending to one. This proves the positive limsup.

For the liminf, take $m = N + r$ with fixed r . Comparing (97) with $\log E_{\kappa_\infty}(N) = -(\log N)^2/(2a) + \mathcal{O}(\log N)$ shows that the quotient tends to zero. $\square$

This is a stronger obstruction than the integer zeros alone: even after replacing the function by the absolute values of its unique local extrema, a constant-amplitude normalization does not exist.

13 A continuum of intermediate boundary rates

The boundary identity also explains why many intermediate effective decay rates appear. Suppose $N = 2^{n-1}$, $z = z_n = o(N)$, and $z_n \rightarrow \infty$. From (13), provided the tail ratio remains bounded away from zero and infinity,

$$\log |\Phi(N + z_n)| = \log |\Phi(z_n)| + n \log \frac{z_n}{N} + o(n). \quad (102)$$

If $\log z_n \sim \alpha \log N$ with $0 \leq \alpha < 1$ and z_n itself follows a typical fixed-phase decay law, then substitution of (30) into (102) gives the quadratic coefficient

$$\log |\Phi(N + z_n)| = -\frac{1 - \alpha + \alpha^2/2}{a}(\log N)^2 + \mathcal{O}(\log N) + o((\log N)^2). \quad (103)$$

The coefficient decreases continuously from $1/a$ at $\alpha = 0$ to $1/(2a)$ as $\alpha \uparrow 1$. This calculation is conditional only on the specified decay law for z_n and makes precise the recursive origin of the broad scatter visible in plots of the extrema.

14 Numerical evaluation of the finite- q spectrum

14.1 Chebyshev collocation

For smooth weights (in particular $q \geq 1$), the Perron root of (21) is efficiently obtained by Chebyshev–Lobatto collocation. Let

$$x_j = \frac{1 - \cos(j\pi/(N-1))}{2}, \quad 0 \leq j < N,$$

interpolate f at the points $x_j/2$ and $(x_j + 1)/2$ by the barycentric formula, and form the matrix representation of $\mathcal{L}_q$. Its leading eigenvalue converges rapidly to ρ_q .

For $q = 1$ the convergence is:

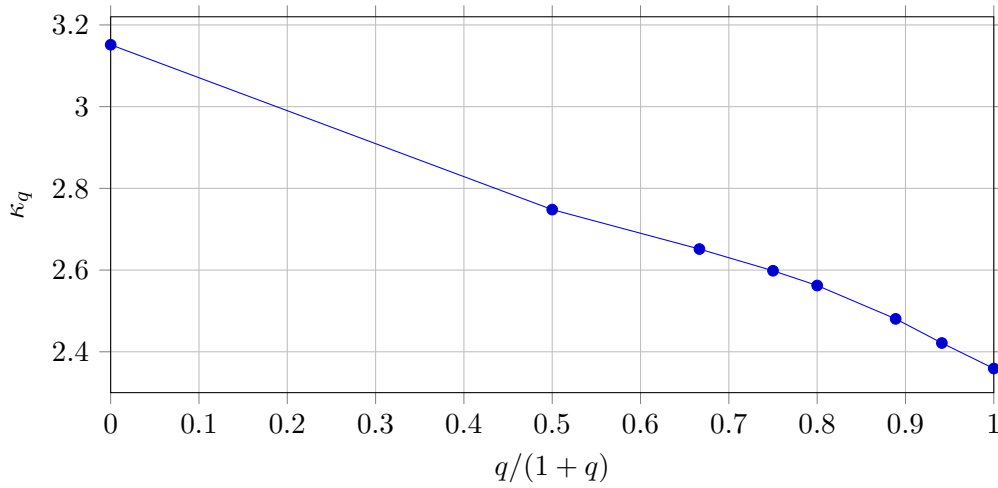
N	leading collocation eigenvalue
8	0.661322639955074
10	0.661322602233874
12	0.661322602061258
16	0.661322602060566
24	0.661322602060565

The rational enclosure in Proposition 9.1 is independent of this numerical convergence and provides a rigorous check on the computed value.

14.2 Sample spectrum

The following values illustrate the interpolation from typical behavior to the sharp uniform envelope. The rows $q = 0, 2, \infty$ are exact; the remaining rows are numerical collocation values.

q	ρ_q	$\Lambda_q = \rho_q^{1/q}$	κ_q
0	–	0.5000000000	3.1514961295
1	0.6613226021	0.6613226021	2.7480700149
2	0.5000000000	0.7071067812	2.6514961295
3	0.3948966324	0.7336593838	2.5983138062
4	0.3201941016	0.7522346457	2.5622414703
8	0.1612444589	0.7960412993	2.4805809434
16	0.0500719357	0.8293247927	2.4214870020
∞	–	0.8660254038	2.3590148791



The monotone curve is not a cosmetic refinement: choosing one row answers one specific notion of “amplitude.” Moving from $q = 0$ to $q = \infty$ changes the power of x by almost 0.8, an effect much larger than any logarithmic correction.

15 Source-by-source reconciliation and correction of the draft

The preceding theorems now permit a precise adjudication of every disputed claim.

15.1 What all four wave-1 documents got right

All four identify the universal factor

$$\exp\left(-\frac{(\log x)^2}{2 \log 2}\right)$$

and reduce the residual behavior to a dyadic lacunary product. They also agree on the integer zero divisor, the Thue–Morse sign law, the existence of one lobe maximum per unit interval, the typical exponent κ_0 , and the sharp pointwise-envelope exponent κ_∞ . These are all correct.

15.2 Why the proposed answers appeared to disagree

Documents 1 and 2 primarily study extremal quantities; Documents 3 and 4 primarily study means. The power of x is determined by the numerator scale being sampled:

$$\frac{1}{2} \quad (q = 0), \quad \rho_1 \quad (q = 1), \quad \frac{1}{\sqrt{2}} \quad (q = 2), \quad \frac{\sqrt{3}}{2} \quad (q = \infty).$$

Substituting these four scales into $\kappa = \frac{1}{2} + \log_2(\pi/\Lambda)$ gives four different, simultaneously correct answers. The old draft's value $\log_2(2\pi)$ is exactly the $q = 2$ row.

Document 2's Lambert term is not a fifth competing power. It refines the $q = \infty$ row when the envelope is anchored at the left endpoint of an entire dyadic annulus. Likewise, Document 4's fixed-offset valleys do not contradict the upper envelope: they show that the sequence of all local maxima has zero liminf on that upper scale.

15.3 The genuine error surface

The only substantive numerical contradiction in the corpus is the LIL constant. Document 1 has $\pi/\sqrt{2}$ with denominator $\sqrt{n \log \log n}$; Document 3 quotes π from [15]. The exact finite- n variance (35) and the explicit martingale decomposition (38) decide the issue in favor of Document 1. The published equation misses the factor $1/2$ in the norm of each real cosine mode.

15.4 The literal Cesaro question

The weak condition from the original question is solved by

$$g(x) = A_1 x^{-\kappa_1} \exp\left(-\frac{(\log x)^2}{2 \log 2}\right), \quad A_1 = 0.091266124131588 \dots$$

It is positive and analytic on $(0, \infty)$, eventually decreasing, and for every fixed m , $(-1)^m g^{(m)}(x) > 0$ for all sufficiently large x (the threshold may depend on m). The normalized ordinary Cesaro average is bounded between positive constants and converges to one along dyadic endpoints. For unrestricted endpoints it converges to a nonconstant mantissa profile. The stabilizing-gauge theorem proves that no ordinary power of x or $\log x$, no slowly varying correction, and no fixed continuous log-periodic modulation can flatten that profile. The logarithmic average does converge for all endpoints.

The strongest possible wording must remain disciplined: the available proof excludes the stabilizing class, not every imaginable analytic monotone function whose relative shape changes at every scale. The comparative audit's broad phrase "unattainable in principle" should therefore be read with this stated hypothesis.

15.5 No universal logarithmic power

At every fixed finite q , Ruelle–Perron–Frobenius asymptotics give a pure exponential in the dyadic level and a nonzero constant; multiplying by $(\log x)^p$ forces the normalized mean to 0 or ∞ unless $p = 0$. At $q = 2$ this is already exact by Parseval. At $q = \infty$ the period-two ray gives an exact constant ratio. Powers of the dyadic level do occur in the exceptional valley formula, but their exponent is the integer zero multiplicity $1 + \nu_2(r + 1)$ and they accompany a different quadratic decay coefficient. They are not a universal half-logarithmic correction.

15.6 Imported analytic inputs and certified parts

The elementary identities, variance computation, Gelfond inequality, Lambert optimization, and fixed-offset valley asymptotic are proved here from scratch. The exact L^2 row is finite Fourier algebra.

The L^1 exponential asymptotic and the enclosure $0.66130 < \rho_1 < 0.66135$ are published inputs from Fouvry–Mauduit and Aistleitner–Hofer–Larcher [16, 15]. The shell-profile limits use the standard Ruelle–Perron–Frobenius spectral gap for the explicit operator (21); this is the one substantial analytic block imported rather than reproved. The independent exact Collatz–Wielandt/Sturm certificate in Section A supplies a machine-checkable enclosure for ρ_1 without relying on floating-point collocation.

16 Conclusion

The Fourier image of Rvachev's up-function has one deterministic core and several genuinely different stochastic or extremal refinements. The exact identity

$$|\Phi(2^n y)| = |\Phi(y)| \pi^{-n} y^{-n} 2^{-n(n+1)/2} \prod_{j=1}^n |\sin(\pi 2^j y)|$$

separates them perfectly. The triangular denominator produces $-(\log x)^2/(2 \log 2)$. The lacunary numerator then selects a pressure level: Lebesgue geometric mean, Perron L^1 mean, Parseval L^2 mean, or maximizing orbit.

The final quantitative dictionary is:

$\kappa_0 = 3.151496129472319 \dots$	typical / geometric mean,
$\kappa_1 = 2.748070014871334 \dots$	arithmetic mean and Cesaro gauge,
$\kappa_2 = 2.651496129472319 \dots$	RMS; the old draft's exponent,
$\kappa_\infty = 2.359014879111741 \dots$	sharp pointwise upper envelope.

Typical logarithms fluctuate on the $\sqrt{\log x \log \log \log x}$ scale with corrected constant $\pi/\sqrt{2}$. Raw dyadic-annulus maxima carry an additional Lambert- W term of order $-(\log \log x)^2/(2 \log 2)$. Fixed-offset lobe maxima after powers of two can instead have the doubled leading coefficient $-(\log x)^2/\log 2$.

For the question that originally motivated the investigation, the positive analytic gauge $A_1 E_{\kappa_1}$ gives the requested two-sided bounded Cesaro normalization and the exact dyadic-endpoint limit. The failure of an unrestricted ordinary limit for all standard stabilizing gauges is structural: the last dyadic shell retains a singular distribution of mass. Logarithmic averaging respects the multiplicative self-similarity and restores an unrestricted limit.

No substantive disagreement among the source documents remains after the norm, endpoint convention, and LIL normalization are made explicit. What looked like one irregular asymptotic is in fact a compact thermodynamic spectrum, supplemented by one Lambert boundary layer and one arithmetic valley mechanism.

A Exact Collatz–Wielandt certificate for ρ_1

The following SymPy script uses only exact rational arithmetic. It constructs the rational function in the proof of Proposition 9.1; Sturm's theorem, invoked by `count_roots`, certifies that the relevant numerators and denominators have no roots on $[0, 1]$. Positivity at $t = 0$ therefore proves positivity throughout the interval.

```
import sympy as sp

t = sp.symbols("t", real=True)
u = 2*t/(1+t**2)
v = (1-t**2)/(1+t**2)
```

```

c = sp.Rational(376189, 10**6)
d = sp.Rational(-69093, 10**6)
e = sp.Rational(15483, 10**6)
F = lambda z: 1 + c*z + d*z**2 + e*z**3

R = sp.cancel((u*F(u) + v*F(v)) / (2*F(2*u*v)))
lo = sp.Rational(66126807, 10**8)
hi = sp.Rational(66134891, 10**8)

for expr in (R - lo, hi - R):
    num, den = map(lambda p: sp.Poly(p, t),
                    sp.together(expr).as_numer_denom())
    assert num.eval(0) > 0 and den.eval(0) > 0
    assert num.count_roots(0, 1) == 0
    assert den.count_roots(0, 1) == 0

print("Certified:", lo, "< rho_1 <", hi)

```

For the high-precision decimal, one may replace the exact certificate by a Chebyshev collocation matrix. The rapid stabilization displayed in [Section 14](#) is typical for this analytic weighted-composition operator.

B Notation cross-reference

Symbol	Meaning
$\Phi = \widehat{\text{up}}$	Fourier transform of Rvachev's up-function
a	$\log 2$
T	doubling map $x \mapsto 2x \pmod{1}$
$s(x)$	$ \sin(\pi x) $
$B_n(y)$	$\prod_{j=1}^n s(T^j y)$
$E_\kappa(x)$	$x^{-\kappa} \exp(-(\log x)^2 / (2 \log 2))$
$W_\kappa(y)$	bounded dyadic phase factor in (10)
$\mathcal{L}_q$	weighted Perron operator (21)
ρ_q	Perron root of $\mathcal{L}_q$
Λ_q	$\rho_q^{1/q}$, the L^q numerator growth per dyadic level
κ_q	sharp power in the normalized L^q shell decay
M_m	unique maximum of $ \Phi $ on $[m, m+1]$

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